

DATA ARCHITECTURE FRAMEWORK

Medallion Data Architecture

BSG: The analytics data flow in 3 easy steps



Bronze: Raw Ingestion



Silver: Cleansed & Conformed



Gold: Business Ready

What You Will Learn Today



Medallion Concept

Understand what the Medallion Architecture is and its core value.



The 3 Stages

Explore how data progresses across Bronze, Silver, and Gold layers.



Stage Breakdown

Detail data formats, transformation rules, and schemas in each stage.



Core Benefits

See how refined data empowers reliable business decision-making.

Multi-Layered Data Refinement



Incremental Value

Each layer increases data quality for confident decisions.



Flexible Ingestion

Collect raw data now and define business usage later.



Early Error Prevention

Progressive cleaning fixes errors early across environments.

The Bronze Layer: Ground Truth



RAW INGESTION LANDING ZONE

Unprocessed landing zone for web APIs, IoT sensors, and transactional databases.

Stores data in raw JSON, CSV, or Parquet files without schema alterations.

As-Is Archival Storage

Preserves original operational data without transformations or filtering.

Permanent Historical Archive

Immutable append-only log annotated with ingestion timestamps for full auditability.

The Silver Layer: Source of Truth



SELF-SERVICE ANALYTICS ENGINE

Cleansed & conformed data foundation serving ad-hoc analysis and self-service analytics.

Validates schema compliance and ensures trusted data quality across enterprise teams.

Thorough Data Cleansing

Filters junk, removes duplicate records, repairs missing values, and fixes nulls.

Format Standardization

Unifies date patterns, splits date/time fields, and enforces column naming standards.

Conformed Business Entities

Joins disparate datasets into unified views of core entities like customers and products.

THE GOLD LAYER

The Gold Layer: Business-Ready Data



EXECUTIVE & BI DECISION LAYER

Highly refined analytical tables optimized for executive dashboards, reporting, and production AI.

Dimensional Star Schemas

Structured star schemas and dimensional models tailored to domain business areas.

Aggregated Reporting Views

Flat tables with pre-calculated aggregations, averages, metrics, and KPI counts.

Business Alignment

Explicit relationships and domain mapping ensure analysts locate data quickly.

BSG data flow



Data Roadmap

- Bronze: 100 sources' tables
- Silver: 40 common model tables
- Gold: 70 business model tables

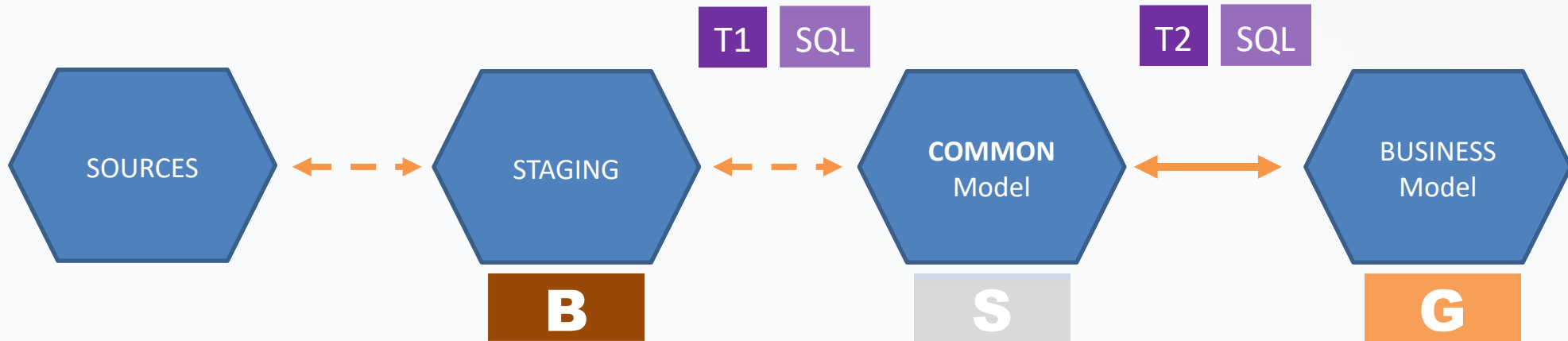


Now What



Is it efficient ? Is it correct ?

The test layer of the Medallion Architecture uses SQL to perform data quality checks and validate the data



BUSINESS BENEFITS

Democratized Data Access



BRONZE CONSUMERS

Data Scientists & AI

Direct access to raw, full-fidelity data for training machine learning models and exploratory research.



SILVER CONSUMERS

Engineers & Analysts

Self-service ad-hoc queries on cleansed, standardized, and conformed enterprise datasets.



GOLD CONSUMERS

Business Executives

Pre-aggregated KPIs and star-schema views for immediate executive decision-making.

ROLE-BASED ACCESS



Zero Interference: Every team accesses their ideal layer seamlessly without cross-department bottlenecks.